Introduction to software engineering

Ing. Josef Pavlíček Ph.D. and collective

List of lectures

1. Lecture - Software engineering
2. Lecture - Software engineering II
3. Lecture - Software process specification
4. Lecture - Software process specification II
5. Lecture - Software modeling
6. Lecture - UML
7. Lecture - System requirements
8. Lecture - System architecture
9. Lecture - Software testing
10. Lecture - User interface testing

List of labs

1. Introduction into the UML and UseCase diagram
2. Sequence diagram
3. Domain diagram
4. Class diagram
5. Activity diagram
6. State diagram
7. Practical work on the Semester work

Semester work conditions

Conditions for Completing the Course:

During the semester, each student will independently work on a semester project. The structure of the semester project is as follows:

- **Title**
- **Problem description** (min. 200 characters) (10 points)
- **Description of actors collaborating in the system** (10 points)
- **Verbal description of use cases min.10 UCs and communication with the system** (10 points)
- **Models** (each worth 10 points):
  - Use Case Diagram for schematic representation of the verbal description of the use case
  - Activity Diagram for describing UseCases scenarios
  - Domain Model
  - Class Diagram
  - Sequence Diagram for Class Diagram (3 pieces)
  - State Diagram of the system for whole system

Semester work Example

More study materials

1. Ian Sommerville. Software Engineering (10th Edition). Pearson, 2015. ISBN: 0133943038, 978-0133943030.
    - Chapter 1 – Introduction to Software engineering
    - Chapter 2 – Socio-technical systems

Authors

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2. prof. Ing. Ladislav Beránek, CSc., MBA (thanks for the theory part)

Contact

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